

Disk-Space Management

1 Basic Idea

Definition

Disk-space management decides how file-system blocks are allocated, freed, tracked, and limited per user.

Main Trade-Off

Small blocks save disk space but reduce performance. Large blocks improve sequential I/O but waste space in small files.

2 File Storage Strategies

Strategy	Advantage	Problem
Contiguous allocation	Fast sequential access.	File growth may require moving the file.
Fixed-size blocks	Handles growth and avoids large moves.	File may be scattered over disk.

Memory Analogy

Contiguous files resemble segmentation; fixed-size disk blocks resemble paging.

3 Block Size Choice

Block size	Good for	Bad for
Small	Space utilization; less waste in last block.	Large-file performance; more seeks and metadata.
Large	Sequential throughput; fewer block accesses.	Small files; more internal fragmentation.

Typical Result

Historically, file systems often used 1 KB to 4 KB blocks. Larger disks make larger blocks more acceptable, but workloads still matter.

4 File-Size Distribution

File size	VU 1984	VU 2005	Web
1 KB or less	48.05	30.91	47.82
4 KB or less	75.31	59.13	70.64
64 KB or less	99.18	90.84	94.80
1 MB or less	100.00	98.87	99.62

Key Observation

Many files are small, but most disk blocks may still be used by a small number of large files.

5 Disk Access Formula

Given Disk

Assume 1 MB per track, rotation time 8.33 msec, and average seek time 5 msec.

Time to Read k Bytes

$$T(k) = 5 + 4.165 + \left(\frac{k}{1,000,000} \right) 8.33 \text{ msec}$$

- 5 msec = average seek time.
- 4.165 msec = half a rotation on average.
- Transfer time is usually tiny for small blocks.

Example

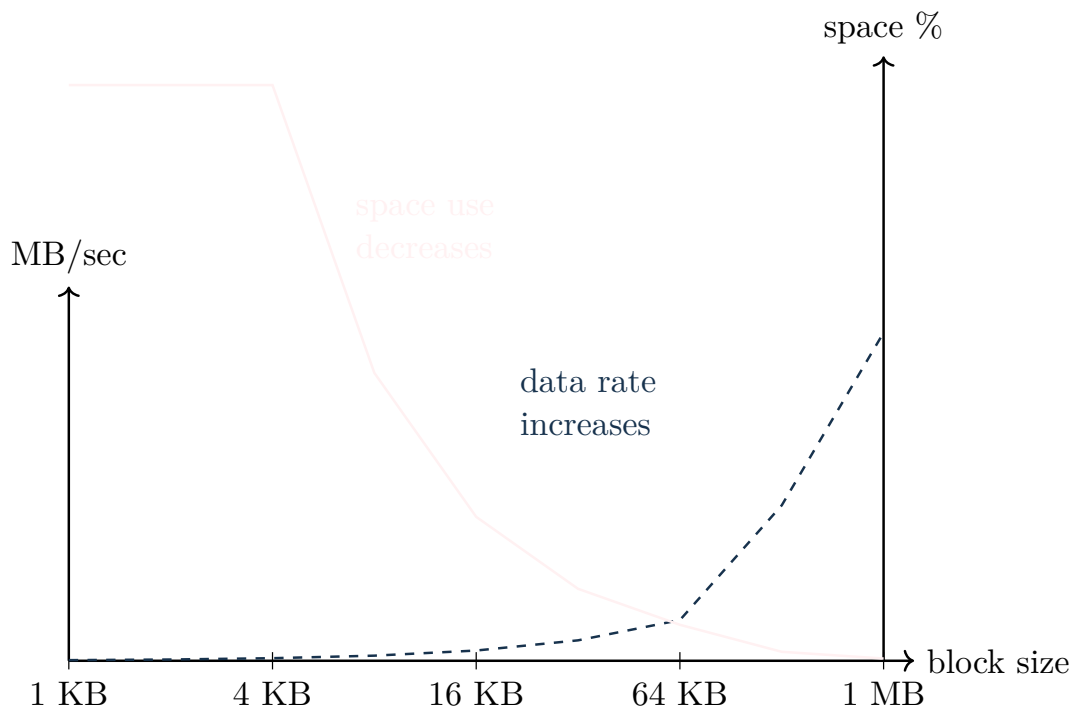
For a 4 KB block, $T \approx 9.20$ msec and the data rate is only about 0.45 MB/sec because seek and rotation dominate.

6 Space Efficiency

$$\text{Efficiency} = \frac{\text{useful file size}}{\text{allocated space}} \times 100\%$$

Block size	Allocated for 4 KB file	Efficiency
1 KB	4 KB	100%
4 KB	4 KB	100%
8 KB	8 KB	50%
16 KB	16 KB	25%
64 KB	64 KB	6.25%

7 Performance vs. Space



8 Tracking Free Blocks

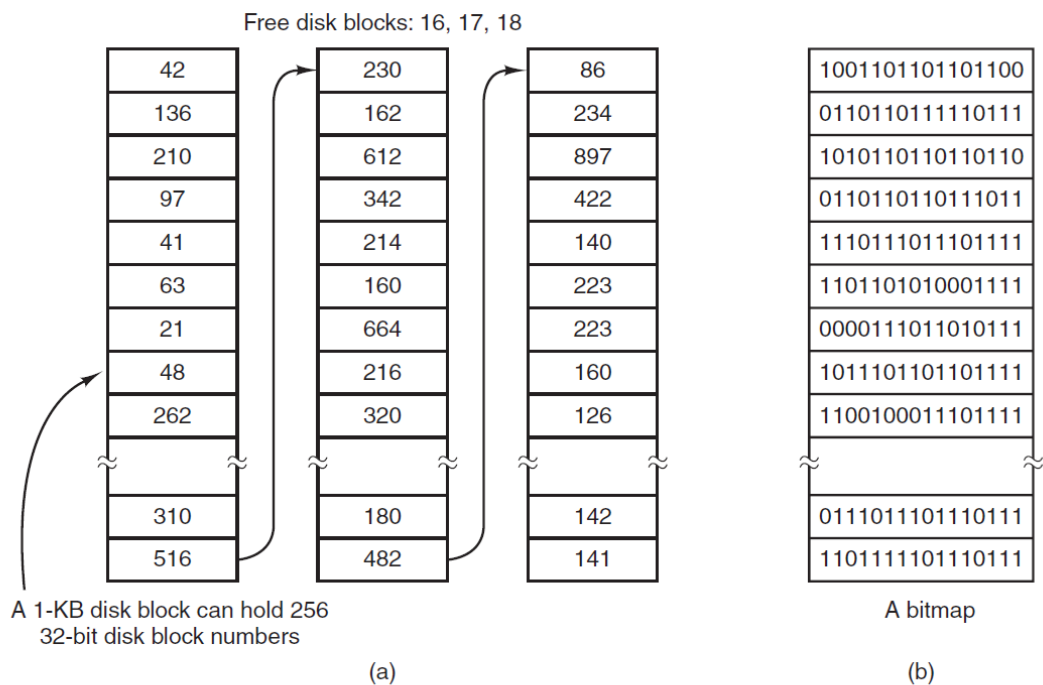


Figure 4-22. (a) Storing the free list on a linked list. (b) A bitmap.

Method	Main idea	Note
Free list	Free blocks store addresses of other free blocks.	Simple; only good when many free blocks need not be searched by position.
Bitmap	One bit per disk block.	Compact; good for finding nearby free blocks.
Runs	Store start block plus count.	Excellent for long free runs; poor if disk is fragmented.

1 TB Example

With 1 billion blocks: bitmap needs about 1 billion bits $\approx 130,000$ blocks of 1 KB. A free list with 255 addresses per block needs about 4 million list blocks.

9 Free-List Pointer Block

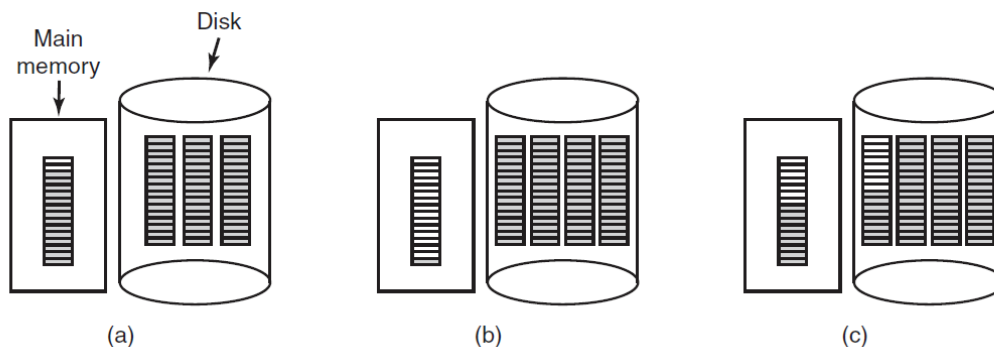


Figure 4-23. (a) An almost-full block of pointers to free disk blocks in memory and three blocks of pointers on disk. (b) Result of freeing a three-block file. (c) An alternative strategy for handling the three free blocks. The shaded entries represent pointers to free disk blocks.

- Keep one block of free-block pointers in main memory.
- File creation removes pointers from it.
- File deletion adds pointers to it.
- If it becomes empty, read another pointer block from disk.
- If it becomes full, write it to disk.

Optimization

Keep the in-memory pointer block about half full. This avoids repeated disk reads and writes when short-lived temporary files are created and deleted.

10 Disk Quotas

Definition

Disk quotas limit how many blocks and files each user may own.

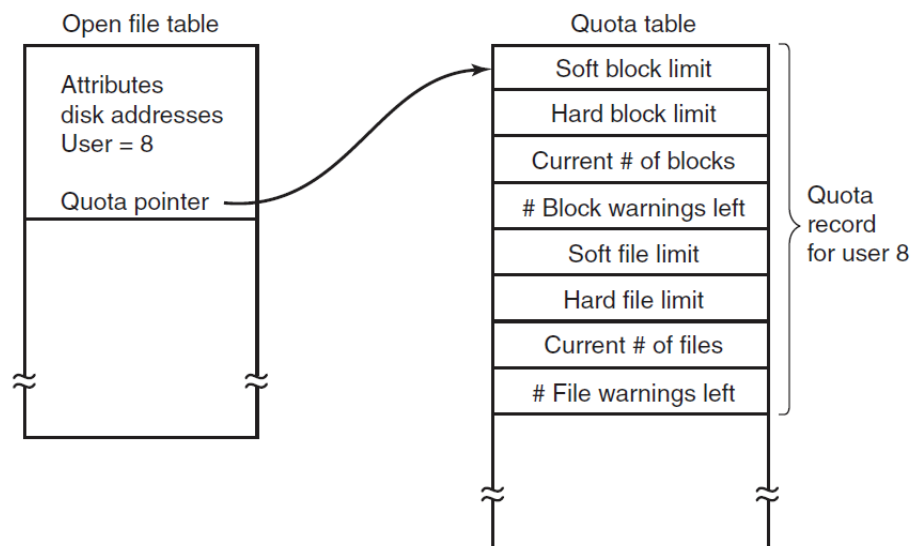


Figure 4-24. Quotas are kept track of on a per-user basis in a quota table.

Quota item	Meaning
Current blocks/files	What the user is using now.
Soft limit	Warning limit; can be exceeded temporarily.
Hard limit	Absolute limit; cannot be exceeded.
Warnings left	Count reduced when user logs in over a soft limit.

How It Works

Each open-file table entry points to the file owner’s quota record. When a block or file is added, the owner’s usage is updated and checked against soft and hard limits.

11 Final Summary

Summary

Disk-space management chooses block size, tracks free blocks, and enforces quotas. Small blocks save space; large blocks improve speed. Free space is commonly tracked by free lists or bitmaps.